DIRECTED CONTAINERS ARE THE CONSTANT-TREE FRAGMENT OF ORGTR, AND THE COMPOSITION OBSTRUCTION LIVES ONLY DOWNSTAIRS

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Abstract. An agent's interface — the queries it may issue and the replies it admits — is a polynomial functor, and a directed container is such an interface equipped with the structure of a small category. Spivak has recently introduced the bicategory **OrgTr**, whose objects are *polynomial trees*: interfaces that reshape themselves as an interaction unfolds. We record a clean positioning of the directed-container programme inside this frontier. First, Spivak's embedding (Proposition 4.6, Corollary 4.8) exhibits the static-interface bicategory **Org** — and with it the fixed-interface world of the DCont ≅ Cof programme — as the full sub-bicategory of **OrgTr** on *constant* (time-invariant) trees. Second, **OrgTr**-composition $(S, \alpha) \# (T, \beta) = (S \times T, \alpha \# \beta)$ is unconditionally total (Proposition 4.3): there is no matching-pair or distributive-law side-condition. We draw the consequence: the degree-two Zappa–Szép obstruction $[\omega] \in H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D})$ that can block a shared-resource composite of two directed containers is intrinsically a phenomenon of the static theory; it has no counterpart at the adaptive-interface level, because the operation it obstructs is not the operation **OrgTr** composes. Making interfaces more general makes composition freer, not more constrained. The sole obstruction internal to **OrgTr** (Remark 3.17, the Zwart–Marsden no-go for $u \triangleleft u$) is of a different kind and is orthogonal to ours.

1. Introduction

The interface reading of Poly. An interactive agent presents an *interface*: a set S of queries it may issue and, for each query $s \in S$, a set $P(s)$ of admissible replies. This is precisely a container, or polynomial functor $\sum_{s \in S} y^{P(s)}$ — the interaction reading of **Poly** due to Spivak and, in the orchestration setting, to Ghani. When the interface additionally carries the structure of a small category — a distinguished "stay" reply at each state, and a way to compose replies into replies — the container is a *directed container* in the sense of Ahman and Uustalu [2], equivalently a comonoid for the substitution product $\triangleleft$ on **Poly**, equivalently a small category. The whole DCont ≅ Cof programme takes this identification seriously: agents with fixed capability-interfaces are directed containers, and composing them is a categorical operation on comonoids.

There is a hypothesis buried in this picture: the interface is *fixed*. The set $P(s)$ of admissible replies to a query does not change as the interaction proceeds. For many systems this is exactly right — a database, a calculator, a well-specified tool. But the systems the Kodamai grant is ultimately aimed at are not like this. An AI agent that *acquires a new capability mid-task* — that learns a tool, reads a manual, or is handed a new sub-agent — has an interface that reshapes itself as a consequence of the interaction. Its set of admissible next moves after step n depends on what happened at steps $1, \ldots, n$. A fixed container cannot model this.

The adaptive frontier: OrgTr. Spivak's recent "Interactions that reshape the interfaces of the interacting parties" [1] builds exactly the setting for reshaping interfaces. Its objects are *polynomial trees*: a root interface, together with, for each (query, reply) pair, a *child* interface — the interface in force after that reply has been given. Iterating, a polynomial tree records an entire coinductive history of interface changes; formally it is an element of the terminal $(u \triangleleft u)$-coalgebra [1, Def. 3.1]. The associated bicategory **OrgTr** organises the state-based interactions between such reshaping

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interfaces. When *every* child interface equals its parent — a *constant tree* — the interface never reshapes, and we are back in the world of fixed interfaces.

This immediately raises a positioning question. If a directed container is a fixed interface with categorical structure, and **OrgTr** is the theory of reshaping interfaces, where does the directed-container programme sit inside **OrgTr**? And, sharper: the directed-container programme has a genuine obstruction — welding two directed containers that share a resource into a single joint container is a Zappa–Szép product, which *need not exist*, its failure measured by a degree-two cohomology class $[\omega] \in H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D})$ [7, 8]. Does this obstruction survive into the more general adaptive setting, or is it an artefact of insisting that interfaces stay fixed?

Results. This note answers both questions. The answers are short, and one of them is a negative that is worth stating plainly. Throughout, we are careful to separate what is Spivak's from what is ours: the two structural theorems below are Spivak's; the *reading* of them — directed containers as the constant-tree fragment, and $[\omega]$ as a strictly downstairs invariant — is our contribution.

- **The embedding (Section 3).** The static-interface bicategory **Org** embeds into **OrgTr** as the full sub-bicategory on constant (time-invariant) trees. This is Spivak's Proposition 4.6 (a fully faithful functor $[p, q]\text{-}\mathsf{Coalg} \to \mathbf{OrgTr}(p, q)$ with image the time-invariant objects) lifted by Corollary 4.8 to a locally fully faithful bifunctor $\mathbf{Org} \to \mathbf{OrgTr}$ [1]. Read through DCont $\cong$ Cof, this exhibits the entire directed-container programme as the *stable-interface fragment* of the adaptive theory.
- **Composition is total (Section 4).** **OrgTr**-composition $(S, \alpha) \,\#\, (T, \beta) = (S \times T, \alpha \,\#\, \beta)$ carries no matching-pair, pullback, or "provided that" side-condition; it is defined directly and closes by coinduction (Spivak, Proposition 4.3 [1]).
- **The obstruction stays downstairs (Observation 4.2).** Consequently the Zappa–Szép obstruction $[\omega]$ does not lift into **OrgTr**'s composition. This is not because the obstruction is trivialised upstairs, but because the operation it obstructs — welding two *objects* along a shared resource — is not the operation **OrgTr** composes, which is total 1-cell composition. The two live at different levels; $[\omega]$ is an invariant of the static welding and has no adaptive-level counterpart.

The moral runs against the usual intuition. Generalising from fixed to reshaping interfaces does not introduce new ways for composition to fail; it composes *more* freely. The obstruction to orchestration is a precise, checkable feature of the *static* composition, and it is exactly the feature that the adaptive setting drops.

For completeness we note (Remark 4.3) that **OrgTr** does contain one genuine obstruction — the Zwart–Marsden no-go blocking $u \triangleleft u$ from being a *monad* (Spivak, Remark 3.17 [1], citing [6]) — but it is a distributive-law non-existence phenomenon, not a cohomological class, and it gates a different construction than ours.

Outline. Section 2 recalls directed containers and the Zappa–Szép obstruction on the one hand, and polynomial trees and **OrgTr** on the other. Section 3 states the constant-tree embedding and the positioning it yields. Section 4 records the totality of **OrgTr**-composition and draws the consequence for $[\omega]$. Section 5 gives the grant-facing reading and an open question.

2. Preliminaries

We keep this section to what the positioning uses. Directed-container basics are in [2]; the **Poly** background is in [4]; polynomial trees and **OrgTr** are in [1], whose numbering we follow.

2.1. **Directed containers and the Zappa–Szép obstruction.** A *container* is a pair (S, P) with $S \in \mathbf{Set}$ and $P \colon S \to \mathbf{Set}$, presenting the polynomial functor $\sum_{s \in S} y^{P(s)}$. A *directed container* equips it with a root section $o \colon \prod_s P(s)$, a reindexing $\downarrow$, and an addition $\oplus$ on positions satisfying the directed-container laws [2, Def. 3.1]. The content of Ahman–Uustalu's theorem is that this is redundant packaging for a familiar object.

Theorem 2.1 (Ahman–Uustalu [2])**.** *Directed containers are equivalent to small categories; equivalently, to comonoids in* $(\mathbf{Poly}, y, \triangleleft)$ *(polynomial comonads on* $\mathbf{Set}$*). Under the equivalence,* S *is the object set of a category* $\mathcal{C}$*,* $P(s)$ *is the set of morphisms out of* s*,* o *is the identity, and* $\oplus$ *is composition. Morphisms of directed containers are cofunctors.*

We write $\mathcal{C}$ for the small category presented by a directed container. When two directed containers $\mathcal{C}, \mathcal{D}$ share a resource — a common set of states through which control passes back and forth — the natural way to run them as a single interleaved system is a *Zappa–Szép product* $\mathcal{C} \bowtie \mathcal{D}$: a category on the joined data in which each side acts on the other, recording who holds control after each handoff. The construction of $\mathcal{C} \bowtie \mathcal{D}$ from a compatible pair of mutual actions (a *matching pair*, equivalently a distributive law of the two directed containers) is Ahman and Uustalu's [3]; we take it from them.

The point that matters here is that $\mathcal{C} \bowtie \mathcal{D}$ *need not exist*: a matching pair assembling the two one-sided actions into an associative whole may fail to close up. This failure is cohomological.

Theorem 2.2 ([7])**.** *The obstruction to closing a Zappa–Szép product of directed containers is a degree-two class* $[\omega] \in H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D})$ *on the skeleton of* $\mathcal{C}$ *with coefficients in the reply functor* $\mathcal{D}$*. The product* $\mathcal{C} \bowtie \mathcal{D}$ *exists if and only if* $[\omega] = 0$*.*

The smallest non-trivial case — a supervisor dispatching to workers through a one-bit turn token — has been computed in full: there $[\omega]$ equals the token-mutation bit, so the joint agent exists exactly when the worker returns control faithfully, with $H^2 \cong \mathbb{F}_2$ [8]. The construction and the token-bit computation are machine-checked in Lean [9].

For this note, all that is needed is the shape of $[\omega]$: it is an invariant of the *welding* operation $\bowtie$, which builds a single new comonoid out of two comonoids that share a resource. Keep this in mind; the contrast in Section 4 turns on it.

2.2. **Polynomial trees and OrgTr.** Fix a universe u of interfaces and let $\mathfrak{c}_{u \triangleleft u}$ denote the cofree $\triangleleft$-comonad, right adjoint to the forgetful functor $U\colon \mathrm{Cmd}(\mathbf{Poly}) \to \mathbf{Poly}$ [1, Prop. 2.2].

Definition 2.3 ([1, Def. 3.1])**.** A *polynomial tree* is an element of the terminal $(u \triangleleft u)$-coalgebra $\mathfrak{c}_{u\triangleleft u}(1)$: a root polynomial $p.\mathsf{root}$ together with, for each position i and direction d, a child tree $p.\mathsf{rest}_{i,d}$. A tree is *constant* (or *time-invariant*) if every child equals the root: $p.\mathsf{rest}_{i,d} = p$ for all i, d.

A constant tree is exactly a fixed interface: the set of admissible next moves never changes. A non-constant tree is an interface that reshapes as the interaction proceeds — the modelling target that fixed containers cannot reach.

Polynomial trees and their morphisms form a category $\mathbf{PolyTr}_U$ [1, Prop. 3.8]. Such a tree-morphism specifies a polynomial map at every node along every interaction path; in practice it is more economically described not node-by-node but by a state set and a transition rule, exactly as a p-coalgebra $(S, \beta\colon S \to p(S))$ compactly encodes an element of the terminal p-coalgebra. Applying this to the internal hom recovers Shapiro and Spivak's bicategory **Org** [5]: a morphism $p \to q$ in **Org** is a $[p, q]$-coalgebra $(S, \beta\colon S \to [p, q](S))$, assigning to each state an action $\mathrm{act}(s) \in \mathbf{Poly}(p, q)$ and an update rule choosing a successor state for each direction (Spivak recalls this at the start of §4 [1]). Parametrising the whole tree of interface changes by a state set in the same way gives **OrgTr**.

Definition 2.4 ([1, Def. 4.1–4.2])**.** For polynomial trees p, q and a state set S, the objects of the hom-category $\mathbf{OrgTr}(p, q)$ are pairs (S, α), where α specifies, coinductively at every node, an action and an update indexed by S; morphisms $(S, \alpha) \to (S', \alpha')$ are state functions $f\colon S \to S'$ intertwining actions and updates at every node.

Objects with singleton state recover the tree-morphisms of **OrgTr**; objects with larger state give compact, state-based descriptions of the same morphisms.

3. The constant-tree embedding

The bridge between the two halves of the preliminaries is Spivak's embedding of the static bicategory into the adaptive one. We state it as it appears in [1] and then read it in directed-container terms.

Proposition 3.1 (Spivak [1, Prop. 4.6, Rem. 4.7])**.** *For polynomials p, q there is a fully faithful functor*

$$[p, q]\text{-}\mathsf{Coalg} \longrightarrow \mathbf{OrgTr}(p, q),$$

sending a coalgebra to the **OrgTr***-object that uses the* same *action-and-update rule at every node. Its image is exactly the* time-invariant *objects of* $\mathbf{OrgTr}(p, q)$*; projection to root-level data gives a retraction* $\mathbf{OrgTr}(p, q) \to [p, q]\text{-}\mathsf{Coalg}$.

Corollary 3.2 (Spivak [1, Cor. 4.8])**.** *The constant-tree assignment $p \mapsto \mathsf{p}$ extends to a locally fully faithful bifunctor* $\mathbf{Org} \to \mathbf{OrgTr}$*: fully faithful on each hom-category by Proposition 3.1, and compatible with composition and identities because both bicategories compose $[p, q]$-coalgebras by the same formula.*

The key clause in Corollary 3.2 is the reason for its proof: the composite of two constant-tree images is computed by the *same* state-pairing-and-action-composing formula as **OrgTr**-composition (Section 4), so the embedding is a genuine bifunctor, not merely a levelwise inclusion.

We now state the reading, which is ours rather than Spivak's.

Observation 3.3 (positioning)**.** A directed container is, by Theorem 2.1, a *fixed* interface — a constant tree — carrying the comonoid structure of a small category. The state-based interactions between fixed interfaces form Shapiro–Spivak's bicategory **Org**, and Corollary 3.2 exhibits **Org** as the *full sub-bicategory of* **OrgTr** *on constant (time-invariant) trees.* The directed-container programme is thus the static-interface corner of the adaptive theory: fixed-capability agents — databases, tools, well-specified sub-agents — are the constant-tree objects; an agent that reshapes its interface mid-task (learns a tool, is handed a new sub-agent) is a genuine non-constant tree, living strictly outside the image of Corollary 3.2.

Spivak states the embedding but does not name the directed-container-as-fragment reading; that framing, and its use to locate the DCont $\cong$ Cof programme as the verified static core of an adaptive theory, is the contribution of this note.

4. Composition is total, so the obstruction stays downstairs

We now record the second structural fact and its consequence for $[\omega]$.

Proposition 4.1 (Spivak [1, Prop. 4.3])**.** **OrgTr** *is a bicategory, with composition*

$$(S, \alpha) \,\#\, (T, \beta) \;\coloneqq\; (S \times T,\ \alpha \,\#\, \beta),$$

where the state set is the bare cartesian product and $\alpha \,\#\, \beta$ composes the two actions node-by-node and pairs the successor states; the identity on p is $(1, \mathrm{id}_p)$. The composite $\alpha \,\#\, \beta$ is defined for all $(S, \alpha), (T, \beta)$: there is no compatibility, matching-pair, or "provided that" side-condition. Associativity and unitality hold at each level of the coinductive tower and hence in the limit.

The proof in [1] *defines* $\alpha \,\#\, \beta$ directly: for $(s, t) \in S \times T$ the composite action is $\mathrm{act}_\alpha(s) \,\#\, \mathrm{act}_\beta(t)$ and the composite update pairs the two side updates; the recursive data is filled coinductively. The only "matching" involved is the trivial domain/codomain one — $\alpha\colon p \to q$ and $\beta\colon q \to r$ meet at the shared tree q, and the two recursive streams land at the same intermediate child automatically. Nothing can obstruct it. The underlying tree composition it rests on is, in Spivak's phrase, "inherited from functions between sets."

This is the crux of the note. There is an apparent tension — downstairs a shared-resource composite of directed containers can fail to exist (Theorem 2.2), yet upstairs composition never fails — and its resolution is exactly the point.

Observation 4.2 (the obstruction is a downstairs invariant)**.** The Zappa–Szép class $[\omega] \in H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D})$ has no counterpart at the level of **OrgTr**-composition, and the apparent tension is not a contradiction, because the two operations are different in kind.

(1) The Zappa–Szép product $\bowtie$ (Theorem 2.2) welds two *objects* — two directed containers, i.e. two comonoids sharing a resource — into a single new object. Building the new comonoid requires a matching pair, and $[\omega]$ measures the failure of that matching pair to close.

(2) **OrgTr**-composition # (Proposition 4.1) composes two 1-*cells* — interactions $p \to q$ and $q \to r$ — into an interaction $p \to r$. This is ordinary composition in a bicategory, and it is total.

OrgTr does not host the welding operation $\bowtie$ at all; its native composition is the total 1-cell composition, and Corollary 3.2 shows the constant-tree embedding respects exactly that total composition. Consequently $[\omega]$ neither lifts into **OrgTr** nor is trivialised there: it remains an invariant of the static welding $\bowtie$, a phenomenon of the directed-container level with no adaptive-level analogue.

The reading is worth stating in plain words: passing from fixed to reshaping interfaces does not create new ways for composition to break. The static theory's one obstruction is a feature of *welding shared-resource objects*, and the adaptive theory simply composes interactions — an operation that was always total. Generality buys freedom here, not friction.

Remark 4.3 (the one **OrgTr**-internal obstruction is orthogonal)*.* **OrgTr** is not obstruction-free; but its one genuine obstruction is of a different kind from ours. Spivak notes [1, Rem. 3.17], citing Zwart and Marsden [6], that $u \triangleleft u$ fails to be a *monad*: its multiplication would require the type-theoretic axiom of choice to hold as an *equality* of sets rather than a bijection, and no such distributive law exists. This is a distributive-law *non-existence* statement, not a cohomological class, and it gates $u \triangleleft u$-monad-ness rather than **OrgTr**-composition — which, using only underlying sets, is unaffected (indeed total, Proposition 4.1). It is therefore orthogonal to $[\omega]$, which classifies non-existence of a Zappa–Szép composite of *static* directed containers. Two obstructions, two levels, two kinds; neither is the other.

5. Discussion

A verified core inside an adaptive frontier. The two facts assemble into a clean map of the orchestration programme. The DCont $\cong$ Cof $\cong$ **Cat** equivalence — proved and machine-checked [2, 9] — is a theory of agents with *fixed* capability-interfaces, and it is exactly the constant-tree fragment of Spivak's **OrgTr** (Observation 3.3). Its one obstruction, the re-entrancy class $[\omega]$, is a precise, checkable property of *static* composition (Observation 4.2). Above it sits the adaptive frontier: agents whose interfaces reshape mid-interaction, where composition is total and the static obstruction does not reach. For the grant this is the honest and useful shape of the story — we hold the verified static core; **OrgTr** is the frontier the AI-mathematician setting will need, and it composes more freely, not less.

An open question. The positioning resolves the *shape* of the relationship but leaves one research question genuinely open. A directed container is a constant tree; an *adaptive* orchestration topology — a migration graph that rewires as a run proceeds — ought to arise as a specific non-constant tree. Does the reshaping of such a tree carry its own holonomy, an adaptive-level analogue of the (G)

closing datum whose failure is $[\omega]$ downstairs? Observation 4.2 shows $[\omega]$ itself does not lift; it does *not* rule out a distinct, genuinely tree-level invariant attached to interface reshaping. Identifying that invariant — or proving there is none — is the natural next target, and it is a prove-session question, not a writing one.

Scope and honesty. To be exact about provenance: Proposition 3.1, Corollary 3.2, Proposition 4.1, and Remark 4.3 are Spivak's, quoted with their numbering from [1]; Theorems 2.1 and 2.2 are Ahman–Uustalu's [2] and the author's [7] respectively, with the base Zappa–Szép construction due to Ahman–Uustalu [3]. What is new here is the synthesis: naming the directed-container programme as the constant-tree fragment (Observation 3.3), and locating $[\omega]$ as a strictly downstairs invariant with no adaptive-level counterpart (Observation 4.2). No new heavy machinery is claimed, and none is needed: the value is the positioning and the honest negative.

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